

Software Requirements Specification (SRS)

CareerPilot: AI-Powered Career Guidance and Learning Roadmap Platform

Version 1.0

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Course

Web and Mobile Application Development

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements of CareerPilot, a web-based platform designed to assist students in planning their career paths through personalized learning roadmaps, progress tracking, and curated educational resources.

This document serves as a reference for project development, testing, and evaluation.

1.2 Scope

CareerPilot is a web application that helps students identify learning paths based on their career goals and current skill levels.

The system allows users to:

- Create and manage accounts.
- Complete career assessments.
- Generate personalized career roadmaps.
- Track learning progress.
- Access learning resources.
- Monitor career development through dashboards.

The project will use React.js for the frontend, Node.js for the primary backend, MongoDB as the database, and Laravel APIs for resource and skill management.

1.3 Definitions and Acronyms

Term	Definition
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SRS	Software Requirements Specification
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API	Application Programming Interface
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CRUD	Create, Read, Update, Delete
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JWT	JSON Web Token
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AI	Artificial Intelligence
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UI	User Interface
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REST	Representational State Transfer
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2. Overall Description

2.1 Product Perspective

CareerPilot is a standalone web application that integrates multiple services to provide personalized career guidance.

The system consists of:

- React Frontend
 - Node.js Backend
 - MongoDB Database
 - Laravel Resource API
 - AI Service Integration
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2.2 Product Functions

The system shall provide:

- User Registration and Authentication
 - Career Assessment
 - Career Roadmap Generation
 - Progress Tracking
 - Resource Recommendations
 - Dashboard Analytics
 - Skill Management
 - Resource Management
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2.3 User Classes

Student/User

Users who:

- Register accounts
- Generate roadmaps
- Track progress
- Access resources

Administrator

Administrators who:

- Manage skills
 - Manage resources
 - Monitor platform content
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2.4 Operating Environment

Frontend

- Google Chrome

- Mozilla Firefox
- Microsoft Edge

Backend

- Node.js Runtime
- Laravel Framework

Database

- MongoDB

Hosting

- Local Development Server
 - Cloud Deployment (Optional)
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3. Functional Requirements

FR-1 User Registration

The system shall allow users to create accounts using:

- Name
- Email
- Password

Inputs

- Name
- Email
- Password

Outputs

- Account Created Successfully
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FR-2 User Login

The system shall authenticate registered users.

Inputs

- Email
- Password

Outputs

- JWT Authentication Token
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FR-3 Profile Management

The system shall allow users to:

- View Profile
 - Edit Profile
 - Update Career Preferences
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FR-4 Career Assessment

The system shall allow users to provide:

- Current Skill Level
 - Interests
 - Preferred Career Path
 - Learning Duration
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FR-5 Roadmap Generation

The system shall generate a personalized roadmap based on assessment data.

Example

Target Career:

Backend Developer

Roadmap:

Phase 1:

Programming Fundamentals

Phase 2:
Node.js & Express

Phase 3:
Databases

Phase 4:
Projects & Deployment

FR-6 Roadmap Management

The system shall allow users to:

- View Roadmaps
 - Update Roadmaps
 - Delete Roadmaps
 - Regenerate Roadmaps
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FR-7 Progress Tracking

The system shall allow users to:

- Mark Skills Complete
 - View Completion Percentage
 - Monitor Learning Progress
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FR-8 Learning Resources

The system shall display:

- Courses
- Documentation
- Tutorials
- Learning Articles

based on selected skills.

FR-9 Skill Management (Laravel Module)

Administrators shall be able to:

- Create Skills
- View Skills
- Update Skills
- Delete Skills

FR-10 Resource Management (Laravel Module)

Administrators shall be able to:

- Create Resources
- View Resources
- Update Resources
- Delete Resources

4. Non-Functional Requirements

NFR-1 Performance

- API responses should be returned within 3 seconds under normal load.

NFR-2 Security

- Passwords shall be encrypted.
- JWT authentication shall be implemented.
- Protected routes shall require authentication.

NFR-3 Availability

- System should be available 24/7 during deployment.

NFR-4 Usability

- Interface should be responsive.
- Navigation should be simple and user-friendly.

NFR-5 Scalability

- The architecture should support future mobile application integration.

NFR-6 Maintainability

- Code should follow modular architecture.
- APIs should be documented.

5. System Architecture

Frontend Layer

React.js

Responsibilities:

- User Interface
- Forms
- Dashboard
- API Consumption

Backend Layer

Node.js + Express

Responsibilities:

- Authentication
- Roadmap Generation

- Progress Tracking
 - User Management
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Laravel API Layer

Responsibilities:

- Skill Management
 - Resource Management
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Database Layer

MongoDB

Responsibilities:

- Store Users
 - Store Roadmaps
 - Store Progress Data
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6. Database Requirements

Users Collection

Attributes:

- User ID
 - Name
 - Email
 - Password
 - Career Goal
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Roadmaps Collection

Attributes:

- Roadmap ID
 - User ID
 - Target Career
 - Learning Phases
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Progress Collection

Attributes:

- Progress ID
 - User ID
 - Skill Name
 - Completion Status
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Skills Table (Laravel)

Attributes:

- Skill ID
 - Skill Name
 - Category
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Resources Table (Laravel)

Attributes:

- Resource ID
 - Title
 - URL
 - Skill ID
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7. Future Enhancements

- Mobile Application
 - Resume Analyzer
 - AI Mentor Chatbot
 - Interview Preparation Module
 - Job Recommendation System
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8. Conclusion

CareerPilot aims to provide students with a structured and personalized approach to career development. By integrating roadmap generation, progress tracking, and learning resource management, the platform will help users make informed learning decisions while demonstrating modern web development practices using React, Node.js, MongoDB, Laravel APIs, and AI-assisted functionality.